

TalentIQ

An AI-Assisted Career Fair Candidate Capture and Review Platform

Prepared for university capstone consideration

Summary

Career fairs create a high-volume, time-constrained recruiting environment. Recruiters may speak with dozens or hundreds of students, often for only a few minutes each. Candidate information is captured through resumes, handwritten notes, memory, or disconnected systems. Because each conversation is documented differently, promising candidates can be difficult to compare after the event, follow-up decisions may take longer than necessary, and important context can be lost.

TalentIQ is a proposed AI-assisted platform that helps recruiters gather concise, consistent candidate information during career fairs and convert it into recruiter-verified candidate snapshots. The system would combine a brief candidate check-in experience, structured recruiter prompts, rapid note capture, AI-supported summarization, and a post-event review dashboard. **TalentIQ** would support decisions about follow-up and next-day interviews while keeping recruiters accountable for every employment-related judgment.

The Problem and Opportunity

- Recruiters have limited time to understand a candidate's qualifications, interests, and communication strengths.
- Notes vary by recruiter and may be incomplete, overly long, or difficult to interpret later.
- Resume details and conversation insights are often stored separately.
- Teams must quickly determine which candidates merit next-day interviews or additional follow-up.
- Manual review makes it difficult to compare candidates consistently across a busy event.

The opportunity is not to automate hiring. It is to improve information quality at the point of contact and reduce the administrative effort required to turn short conversations into timely, well-supported next steps. The goal is to help recruiters spend less time organizing information and more time engaging with candidates.

Proposed Solution

TalentIQ would provide a lightweight web application optimized for laptops, tablets, and mobile devices used at a career fair. Candidate-entered information and recruiter-entered observations would be combined into a concise, standardized record. AI would help organize and summarize candidate-provided information and recruiter observations into a structured draft profile. Recruiters would review, edit, approve, or reject every AI-generated summary before it becomes part of the candidate record.

Example Candidate Data Model

To support consistency across recruiters and events, the prototype should maintain a structured candidate record. The following fields are representative and **may be refined by the student team during requirements discovery**.

Candidate Information

- Candidate ID
- First Name
- Last Name
- Preferred Name (optional)
- Email Address
- Phone Number (optional)
- University
- Degree Program
- Major
- Graduation Date
- GPA (optional)
- Resume Upload

Career Interests

- Desired Internship or Job Function
- Areas of Technical Interest
- Preferred Work Location(s)
- Relevant Skills
- Relevant Coursework
- Project Experience

Work Authorization

- Work authorization information (optional)

Recruiter Observations

- Recruiting Event
- Recruiter Name
- Conversation Notes
- Areas of Interest Discussed
- Follow-up Questions
- Recommended Next Steps
- Candidate Questions

AI-Assisted Summary

- Candidate Snapshot
- Key Skills Identified
- Relevant Experience Identified
- Missing Information Flags
- Source Citations/Traceability References

Recruiting Workflow

- Record Status (New, Reviewed, Follow-Up, Interview Requested, Closed)
- Recruiter Approval Status
- Approval Timestamp
- Last Updated Date

Core MVP Capabilities

Capability	MVP Requirement
Candidate intake	QR-accessible, mobile-friendly form with a deliberately small number of required fields.
Recruiter capture	Fast structured prompts, tags, and short free-text notes designed for a three-to-five-minute conversation.
AI summarization	A standardized candidate snapshot grounded only in supplied information, with missing-data indicators.
Human verification	Edit-and-approve workflow, visible source notes, and a record of who approved the summary.
Review dashboard	Search, filters, side-by-side comparison, status assignment, and concise follow-up views.
Measurement	Instrumentation that supports a controlled comparison of the traditional and TalentIQ-supported processes.

Role of Artificial Intelligence

AI should assist with information organization, **not employment decision-making**.

Appropriate AI functions include:

- Draft candidate summaries grounded in candidate and recruiter-provided information.
- Identification of missing or incomplete candidate information.
- Natural-language search across recruiter-approved candidate records.
- Citation-based summaries that link generated content to supporting source information.
 - *Example: Demonstrated experience with Python and React. [Resume]*
- Voice-to-text recruiter note capture and organization.

Optional AI Enhancements

Students are encouraged to innovate beyond the minimum requirements by proposing additional AI-powered features that enhance the recruiting experience

Example:

- **Interview question suggestions** - Based on the candidate's resume and responses, AI suggests follow-up questions.
 - *Example: Candidate mentioned machine learning research. → Suggested questions: What was your contribution to the project?*

The AI should not:

- Automatically reject, advance, rank, or score candidates for employment.
- Infer personality, emotion, disability, race, ethnicity, age, gender, religion, or any other protected or sensitive characteristic.
- Generate qualifications or conclusions that are not supported by the source information.
- Use candidate data to train an external model without explicit authorization.

Accessibility and Inclusive Design

Accessibility is a functional requirement, not an optional enhancement. The prototype should be designed and evaluated against relevant WCAG 2.2 Level AA principles.

At minimum, the solution should support:

- keyboard-only use
- clear focus states
- properly labeled form controls
- sufficient color contrast
- readable error messages
- responsive zoom and reflow

Privacy, Security, and Responsible Use

Use synthetic data during development and demonstration unless the sponsor explicitly approves another data source. Any approved data must be handled in accordance with applicable privacy, security, and institutional requirements

Project Scope

In Scope

- Responsive candidate intake and recruiter capture experiences.
- Candidate record storage using synthetic sample data.
- AI-generated draft summaries with recruiter verification.
- Post-event filtering, comparison, status assignment, and export.
- Accessibility evaluation and remediation of identified issues.
- A controlled study comparing the traditional process with the prototype-supported process.

Out of Scope for the MVP

- Production integration with an applicant tracking system, HR platform, or enterprise identity provider.
- A production deployment containing real candidate data.

Expected Deliverables

- **Working application prototype.** A functional end-to-end MVP with representative data and a documented setup process.
- **AI implementation and evaluation.** Prompt or model design, grounding approach, failure handling, and tests for unsupported or inaccurate summaries.
- **Accessibility evidence.** An accessibility checklist, testing results, known limitations, and remediation completed during the project.
- **Comparative process study.** Results showing how the traditional workflow compares with the TalentIQ-supported workflow.
- **Final demonstration and recommendations.** A sponsor presentation, application demonstration, lessons learned, and a practical roadmap for future development.

Evaluation Plan: Traditional vs. TalentIQ

The student team should create a simulated career-fair dataset and ask representative users to complete the same review task using two methods: the current manual approach and the TalentIQ-supported approach. The study should evaluate both efficiency and information quality.

Measure	What It Tests
Capture time	Average time required to create a usable candidate record.
Review time	Time required to identify candidates for a defined follow-up action.
Completeness	Percentage of required information captured across candidate records.
Consistency	Degree to which records follow the same structure and can be compared.
Summary accuracy	Percentage of AI summary statements supported by source information.
Correction effort	Number and type of edits recruiters make before approving a summary.
Decision confidence	Recruiter-reported confidence that follow-up choices are supported by adequate information.
Usability and accessibility	Task completion, error rate, user feedback, and results of accessibility testing.

Definition of Success

TalentIQ offers a meaningful and achievable capstone challenge grounded in a real recruiting need. It asks students to build more than a generic AI demonstration. The team must design for a fast-paced recruiting environment, preserve human accountability, protect candidate information, meet accessibility expectations, and demonstrate whether the solution improves the recruiting process.

Potential benefits include improved recruiter efficiency, more consistent candidate documentation, faster follow-up decisions, and an enhanced recruiting experience for both students and recruiters.

The capstone will be considered successful if the team demonstrates that the prototype:

- Supports the complete workflow from candidate check-in through a recruiter-approved follow-up status.
- Produces concise candidate summaries whose statements are traceable to candidate-provided information or recruiter notes.
- Reduces post-event review effort or improves record completeness and consistency compared with the manual process.
- Meets the agreed accessibility acceptance criteria and documents any remaining limitations.

Supporting Information:

Tech Stack

The university team may select the technology stack, AI model, database, and hosting approach most appropriate for the project objectives.

Sponsor Support

Carl Pegue, Manager, Engineering and Technology, J.B. Hunt Transport, Inc.
carl.pegue@jbhunt.com

The sponsor will provide:

- Business context and requirements
- Periodic feedback sessions
- Clarification of recruiting workflows
- Access to representative synthetic datasets and sample artifacts where appropriate

Primary Users

- University Recruiters
- Hiring Managers attending career fairs
- Recruiting Coordinators